

Privacy Preserving Link Prediction (PPLP)

Kaleb Kim, Wiam Skarki, Carson Whitehouse,
Jonah Lorenzo, Kent Manion, Aidan Bugayong

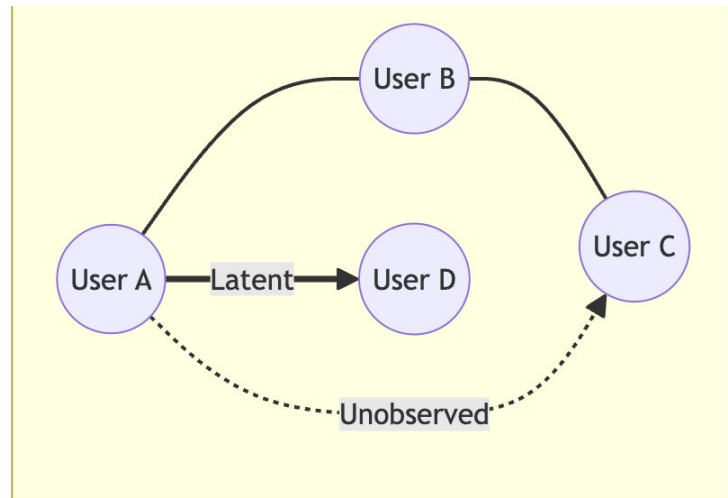
Our Goal

To build a Python library that allows anyone to apply **privacy preserving link prediction**.

Link Prediction

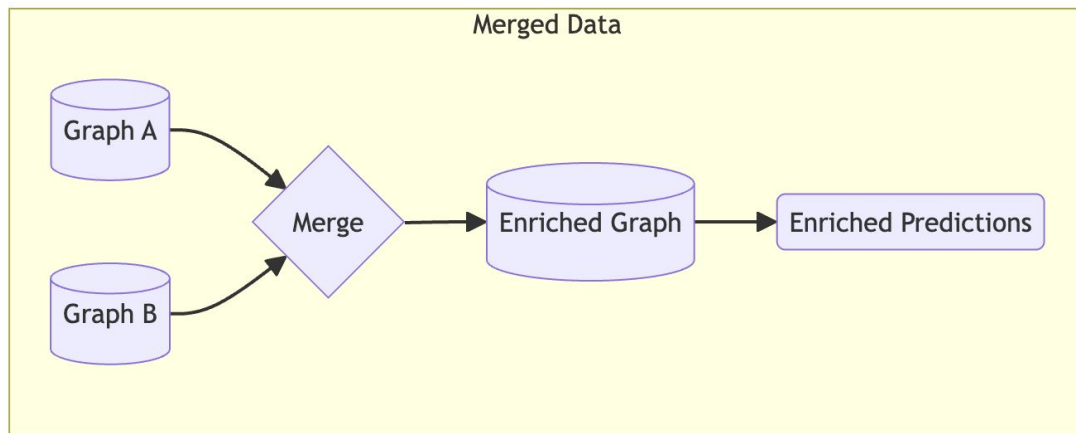
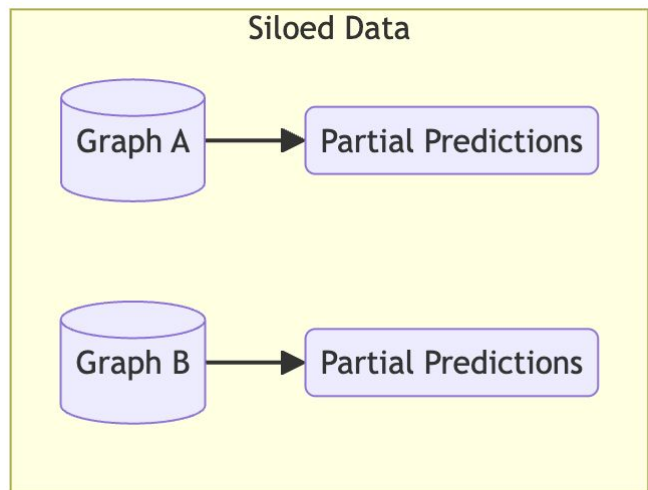
Link Prediction

- Link prediction helps discover unobserved or latent connections between nodes in a graph
- Data holders rank the likelihood of new connections forming overtime
- Useful for social networks, e-commerce, telecomms, bioinformatics...

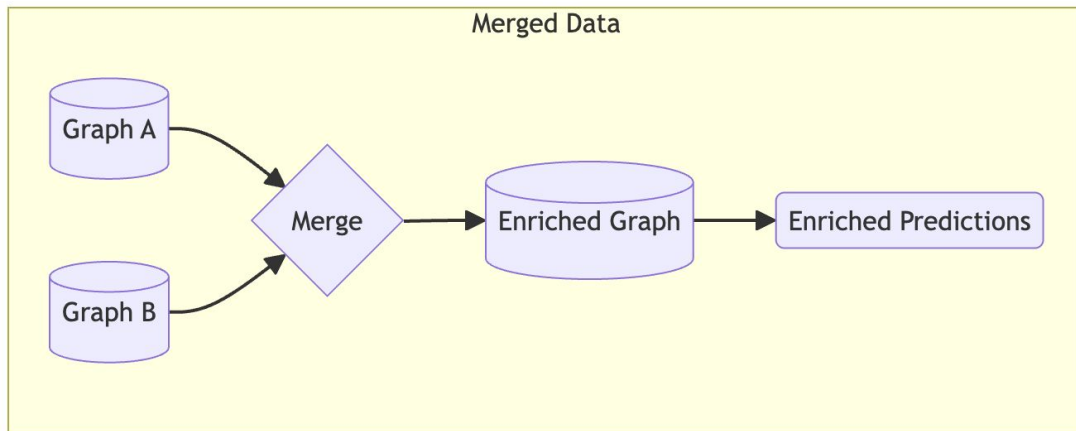


Link Prediction

- Link prediction is typically done on a single local graph
- Link prediction can be more accurate if we merge two or more graph databases that include similar information (Distributed Link Prediction)



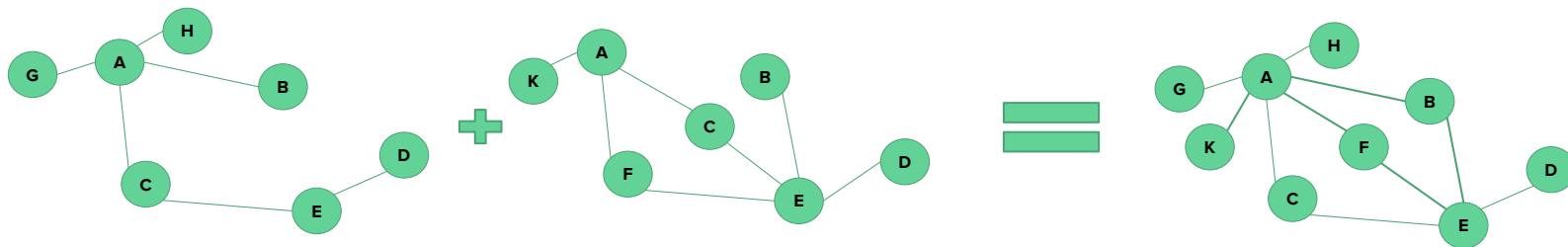
Distributed Link Prediction



- Two parties can utilize the connections in their combined graph to provide more accurate link predictions
- In other cases, distributed link prediction allows link prediction between nodes based on other attributes

Distributed Link Prediction

- In some cases, collaboration is mutually beneficial to contributing parties
- In other cases, the second party is paid to participate
- Distributed Link Prediction results in privacy concerns since it implies combining two or more different graph databases!
 - Identity Disclosure
 - Link Disclosure
 - Attribute Disclosure



Common Neighbors Measure

$$CN(A, B) = |A \cap B|$$

- Computes the number of common neighbors between two parties
- More neighbors = Higher link likelihood
- Fails to account for the relative amount of neighbors

Jaccard Measure

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

- Computation of the relative amount of neighbors in common
- This improves on the common neighbors measure

Adamic-Adar Measure

$$A(x, y) = \sum_{u \in N(x) \cap N(y)} \frac{1}{\log |N(u)|}$$

- $N(x) \cap N(y)$ represents the set of common neighbors of x and y
- $|N(u)|$ represents the degree of node u
- The sum of the log of the intersection of the neighbors of two nodes
- Two-hop similarity helps to yield better results

Privacy Preserving Link Prediction (PPLP)

Privacy preserving link prediction allows multiple data holders to collaboratively forecast unobserved or latent connections between nodes **without explicitly revealing what each party knows.**

Use Cases I

- **Social Networks:** Privacy-First recommendations
 - Discovering friend suggestions between users based on structural similarity (Think of # Mutual Connections on Instagram!)
 - Does not expose the full contact list of either party
- **Telecommunications:** Targeted Advertising
 - Finding target audience members similar to an existing set
 - Link prediction on phone networks while keeping the actual call/link records private

Use Cases II

- **E-Commerce:** Collaborative Filtering
 - Recommending products to a user based on similarities between them and others' purchasing
 - Does not use a central server seeing individual transaction histories
- **Bioinformatics:** Secure EHR Research
 - Merging Electronic Health Records (EHR) with clustering graphs to predict disease spread or patient diagnoses
 - Does not reveal sensitive protected health information (PHI).

Private Set Intersection

- PPLP heavily uses Private Set Intersection (PSI) protocols
- There are many different algorithms that leverage both Oblivious Transfer (OT) and Fully Homomorphic Encryption (FHE), but OT is bottlenecked by communication complexity

Homomorphic Encryption

Homomorphic encryption allows for direct operations on encrypted data without knowing the contents of the encrypted data. This allows us to do Private Set Intersection as shown below:

$$P(X) = r \cdot (X - x_0) \cdot \dots \cdot (X - x_9).$$

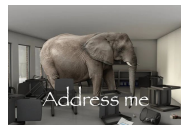
Where:

X exists in Graph 1 and Graph 2 iff $P(X) = 0$

- X is a fully homomorphically encrypted node in Graph 1
- x_i is node i in Graph 2
- r is a random constant

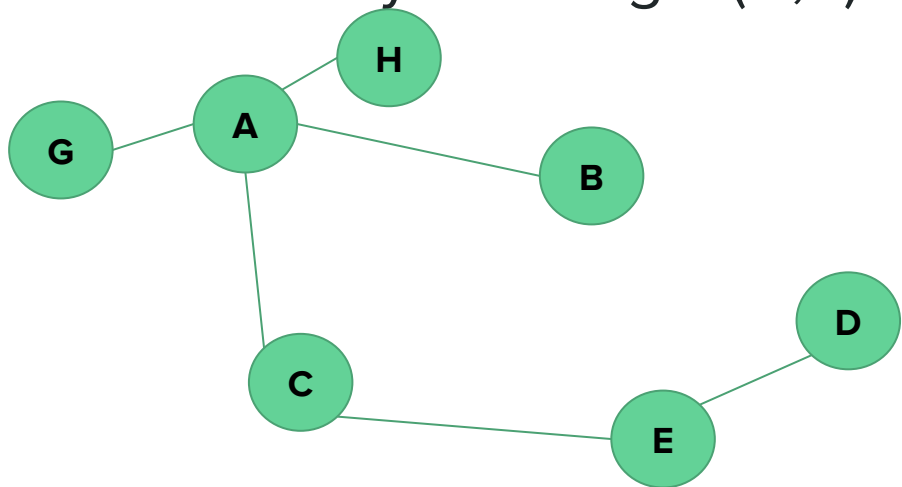
~~Demirag~~ Ayday et al. (2022): Privacy-Preserving Link Prediction

- Two parties (Graph 1, Graph 2) compute Common Neighbours across their joint graphs without revealing their graph to each other (link prediction)
- $CN = local1 + local2 + crossover1 + crossover2 - overlap$
 - Computed using 3 PSI calls
- Leaks some intermediate values and proposes a heavier homomorphic implementation for zero leakage

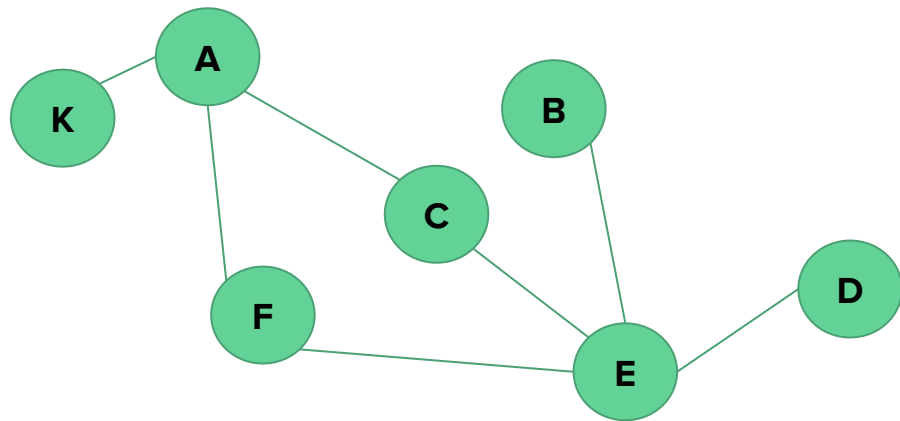


Higher Level Example

Subset only Looking $N(A,E)$

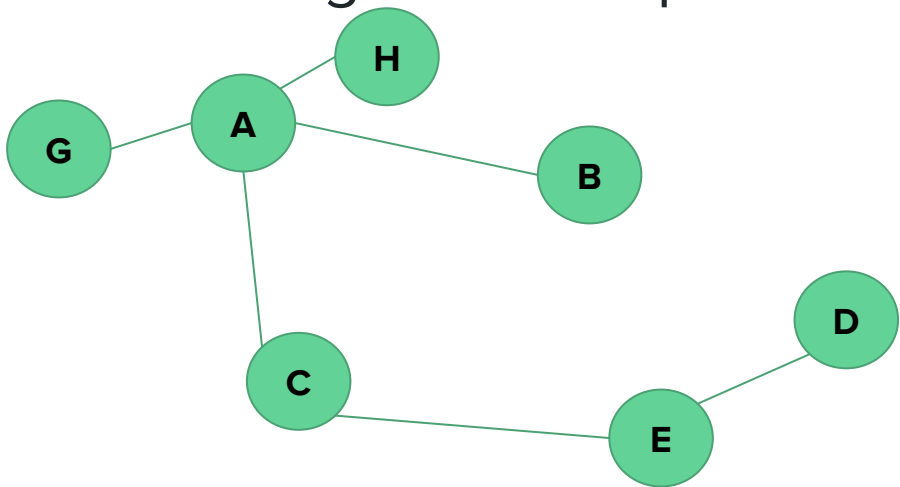


Alice



Bob

Searching Local Graph

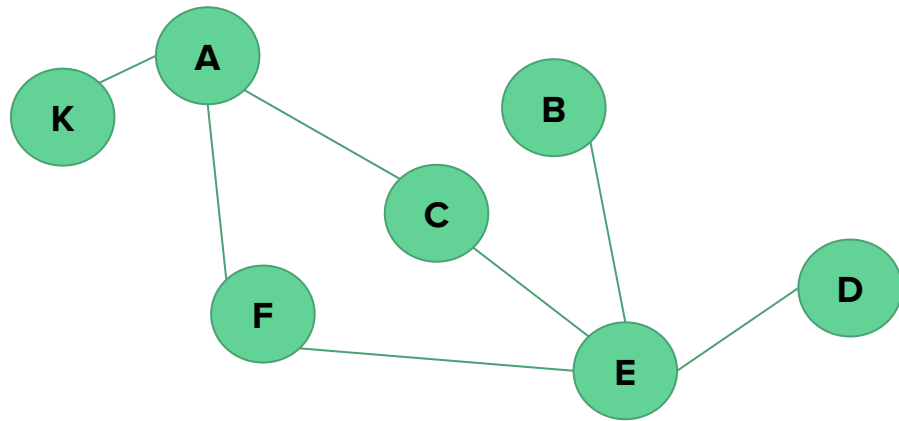


Alice

$$N_A(A) = \{H, G, B, C\}$$

$$N_A(E) = \{D, C\}$$

$$N_A(A) \cap N_A(E) = I_A(A, E) = \{C\}$$



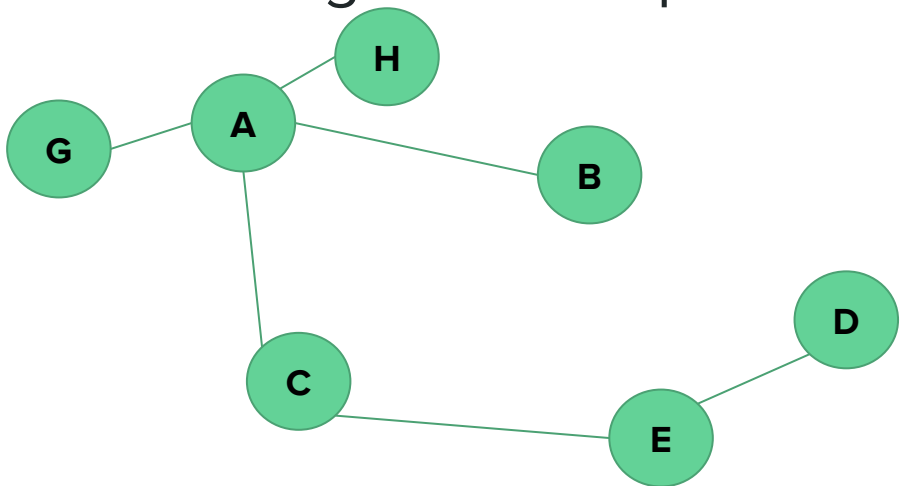
Bob

$$N_B(A) = \{K, F, C\}$$

$$N_B(E) = \{B, C, F, D\}$$

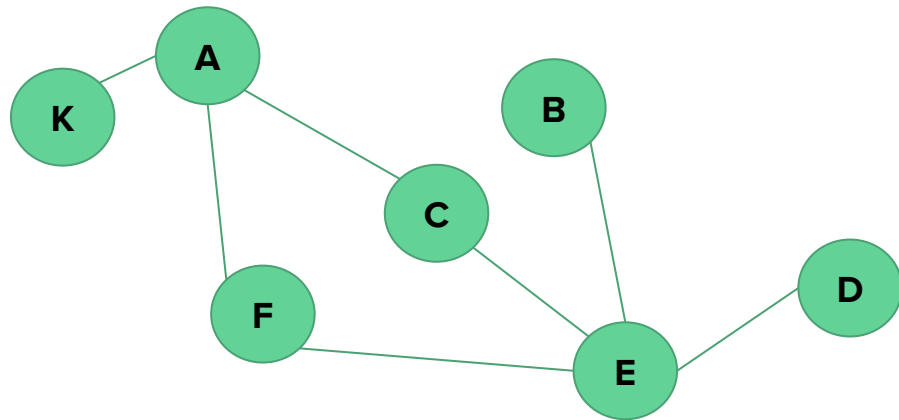
$$N_B(A) \cap N_B(E) = I_B(A, E) = \{C, F\}$$

Searching Local Graph



Alice

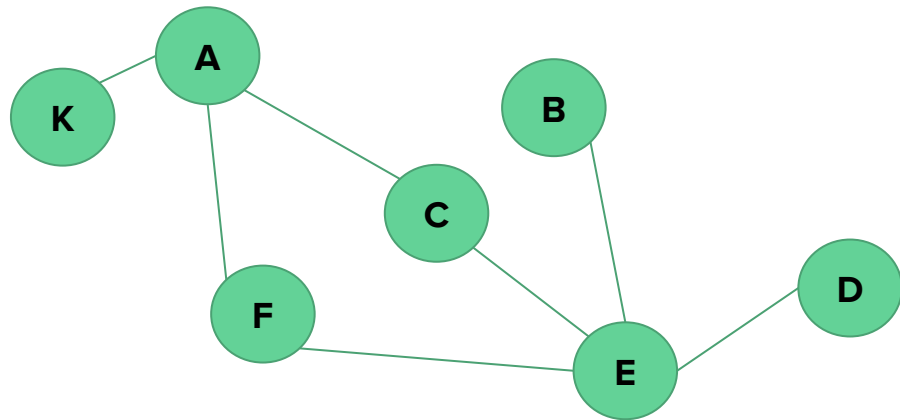
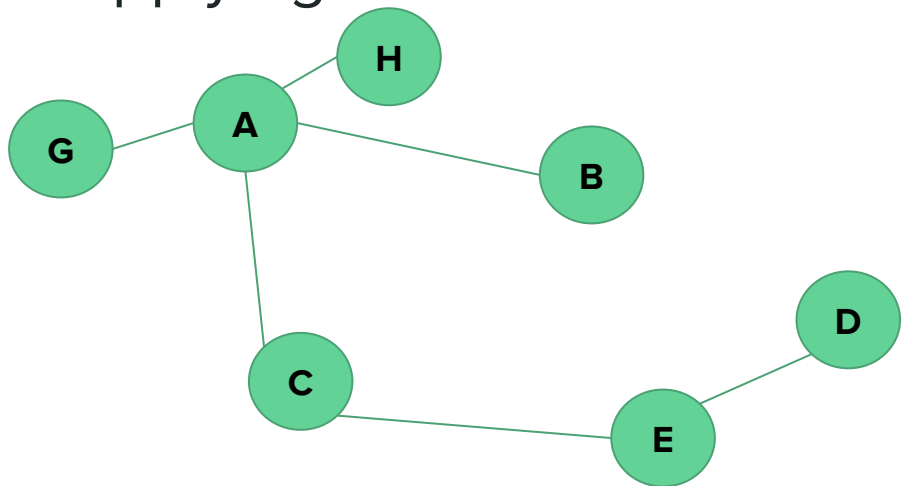
$$\begin{aligned}N_A(A) &= \{H, G, B, C\} \\N_A(E) &= \{D, C\} \\N_A(A) \cap N_A(E) &= I_A(A, E) = \{C\} \\N_A(A) - I_A(A, E) &= \{H, G, B\} \\N_A(E) - I_A(A, E) &= \{D\}\end{aligned}$$



Bob

$$\begin{aligned}N_B(A) &= \{K, F, C\} \\N_B(E) &= \{B, C, F, D\} \\N_B(A) \cap N_B(E) &= I_B(A, E) = \{C, F\} \\N_B(A) - I_B(A, E) &= \{K\} \\N_B(E) - I_B(A, E) &= \{B, D\}\end{aligned}$$

Applying PSI



Alice

$$\begin{aligned} N_A(A) \cap N_A(E) &= I_A(A, E) = \{C\} \\ N_A(A-E) &= \{H, G, B\} \\ N_A(E-A) &= \{D\} \end{aligned}$$

$$\begin{aligned} N_B(A) \cap N_B(E) &= I_B(A, E) = \{C, F\} \\ N_B(E-A) &= \{B, D\} \\ N_B(A-E) &= \{K\} \end{aligned}$$

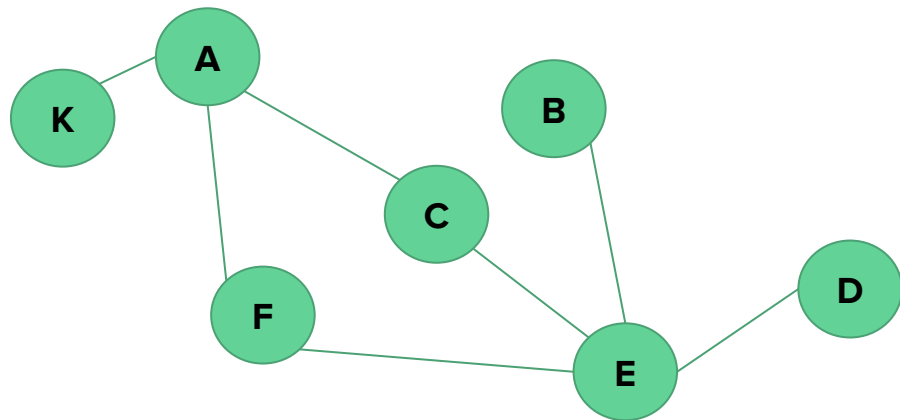
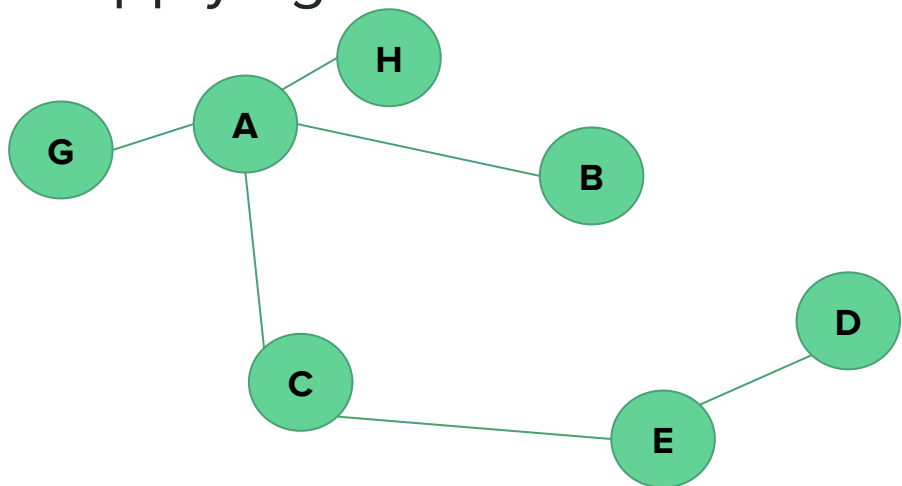


Bob

PSI

$$I_A \cap I_B = I_{A+B} = \{C\}$$

Applying PSI



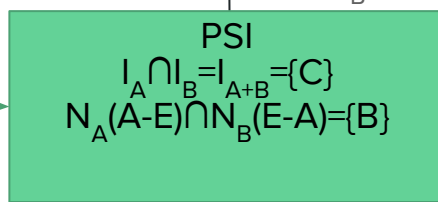
Alice

$$\begin{aligned}
 N_A(A) \cap N_A(E) &= I_A(A,E) = \{C\} \\
 N_A(A-E) &= \{H, G, B\} \\
 N_A(E-A) &= \{D\}
 \end{aligned}$$

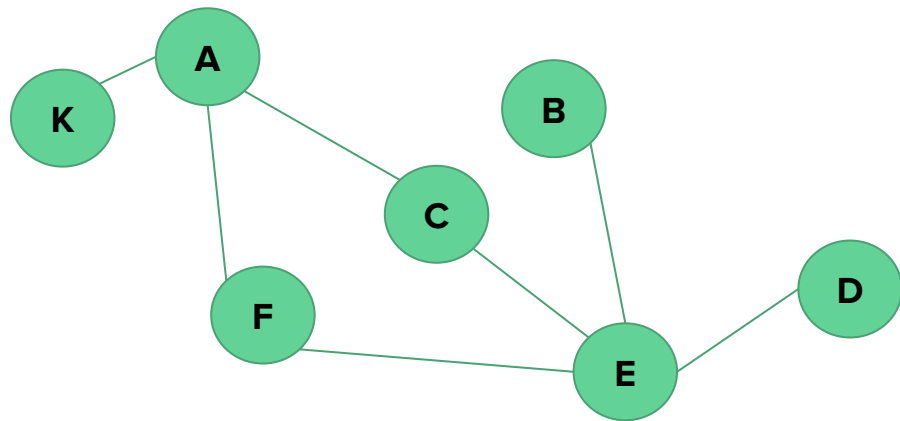
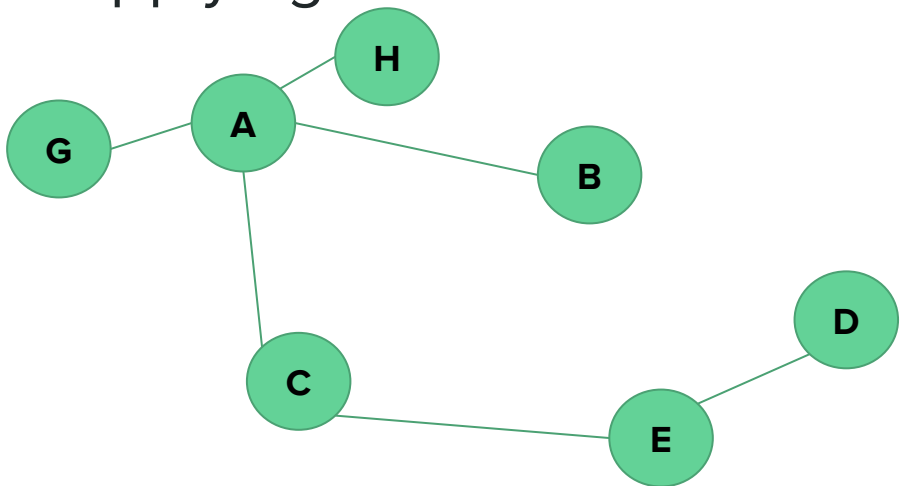
$$\begin{aligned}
 N_B(A) \cap N_B(E) &= I_B(A,E) = \{C, F\} \\
 N_B(E-A) &= \{B, D\} \\
 N_B(A-E) &= \{K\}
 \end{aligned}$$



Bob



Applying PSI



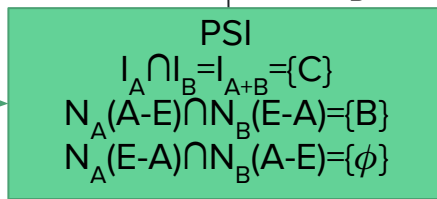
Alice

$$\begin{aligned}N_A(A) \cap N_A(E) &= I_A(A, E) = \{C\} \\N_A(A-E) &= \{H, G, B\} \\N_A(E-A) &= \{D\}\end{aligned}$$

$$\begin{aligned}N_B(A) \cap N_B(E) &= I_B(A, E) = \{C, F\} \\N_B(E-A) &= \{B, D\} \\N_B(A-E) &= \{K\}\end{aligned}$$

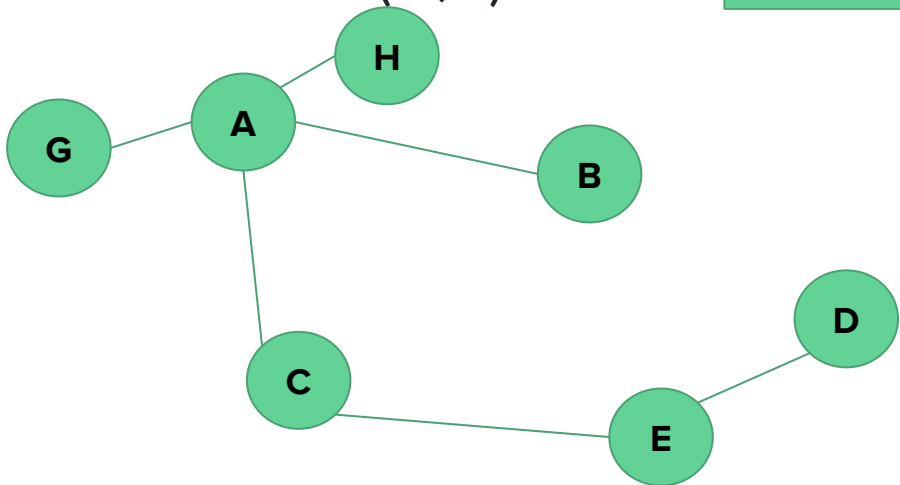


Bob



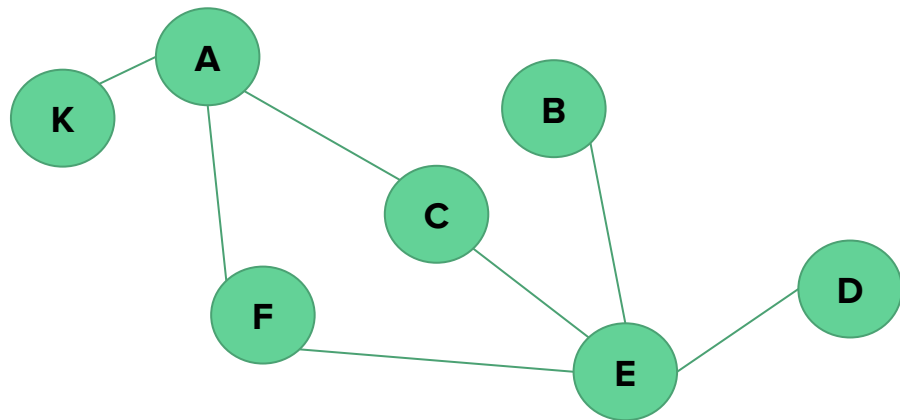
Size of $CN(A,E)$

$$|CN(A,E)| = |I_A| + |I_B| - |I_{A+B}| + |\{B\}| + |\{\phi\}| = 3$$



Alice

$N_A(A) = \{H, G, B, C\}$
 $N_A(E) = \{D, C\}$
 $I_A(A, E) = \{C\}$



Bob

$N_B(A) = \{K, F, C\}$
 $N_B(E) = \{B, C, F, D\}$
 $I_B(A, E) = \{C, F\}$

PSI

$$I_A \cap I_B = I_{A+B} = \{C\}$$

$$N_A(A-E) \cap N_B(E-A) = \{B\}$$

$$N_A(E-A) \cap N_B(A-E) = \{\phi\}$$

Previous Work Done

Liben-Nowell & Kleinberg (2004): The Link Prediction Problem

- Defines link prediction as a formal problem:
- Given a graph snapshot at time t , predict which edges will appear by time t' using only network topology and no node attributes.

Gentry (2009): Fully Homomorphic Encryption Using Ideal Lattices

- Defines FHE as a viable scheme
 - Uses the Bootstrapping technique
 - Solves the noise problem
- Constructs the scheme using ideal lattices

Chen, Laine & Rindal (2017): Fast PSI from Homomorphic Encryption

- A high-performance PSI protocol using Fully Homomorphic Encryption (FHE)
- Communication complexity is $O(N_{\text{small}} \cdot \log N_{\text{large}}) \rightarrow O(n \log n)$
- While FHE makes it asymptotically efficient, it is not efficient in practice and still leaks minor information

Ling et. al (2025): Ultra-Fast Private Set Intersection From Efficient Oblivious Key-Value Stores

- An ultra-fast PSI protocol built upon a novel, bucket-based Oblivious Key-Value Store (OKVS) and Vector Oblivious Linear Evaluation (VOLE)
- Communication complexity is $O(n)$
- Minimizes network communication overhead compared to previous leading protocols: 30% → 1% redundancy
- Used for our implementation

Project Details

We'll build a Python library that allows anyone to apply **privacy preserving link prediction**.

Implementation

- Backend
 - Ultra-Fast PSI protocol from Ling et. al, open-sourced
 - Uses C++ lib in Ling et. al for $O(n)$ communication complexity
- Library Interface
 - Bridges the C++ backend to a user-friendly Python interface
- Web App Interface
 - Streamlit dashboard for uploading graphs and predict links
 - Interactive graph rendering

ref.

- [Privacy Preserving Link Prediction](#) (Ayday)
- [Fast Private Set Intersection from Homomorphic Encryption \(PDF\)](#)
- [Feather: Lightweight Multi-party Updatable Delegated Private Set Intersection](#)
- [The Link Prediction Problem for Social Networks](#)
- [Lucidchart](#)

<https://github.com/ShallMate/fastpsi> – [Ultra-Fast Private Set Intersection \(PDF\)](#)